Response to Quentin's Suggestion

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Quentin is questioning this equality:

$$Ep = \int_{\underline{p}}^{\tilde{p}} (1 - H(p))\, dp$$

and argues that this is not true for two reasons:

1. the expecation is conditioned on the event $p \leq \tilde{p}$
2. the domain of integration $\left[\underline{p}, \tilde{p}\right]$ is not the support of the random variable p, which is $\left[\underline{p}, B\right]$, where B is the upper bound for the price that firms offer.

To answer this question with one line: <u>the question is valid, but it is not a concern in this particular case since in this case $\tilde{p} = B$ in the equilibrium as no firm will offer price higher than the consumer's reservation price</u>. Still, the equality is not fully correct. If we do the math for the right hand side using the integration by parts formula and the facts $H(\tilde{p}) = 1$ and $H\left(\underline{p}\right) = 0$, we would have the following

$$\begin{aligned}
\int_{\underline{p}}^{\tilde{p}} (1 - H(p))\, dp &= (1 - H(p))\, p|_{\underline{p}}^{\tilde{p}} - \int_{\underline{p}}^{\tilde{p}} p d(1 - H(p)) \\
&= 0 - \underline{p} + \underbrace{\int_{\underline{p}}^{\tilde{p}} p dH(p)}_{\equiv E[p]} \\
&= E[p] - \underline{p}.
\end{aligned}$$

In general, $E[p] \geq \int_{\underline{p}}^{\tilde{p}} (1 - H(p))\, dp$, and the equality holds only when $\underline{p} = 0$.

Nevertheless, the key equality $\tilde{p} = c + Ep$ that we would like to show in the chapter 6 is correct. If we use the definition of unconditional expecation $Ep = \int_{\underline{p}}^{\tilde{p}} p dH(p)$, we can get the following,

$$\begin{aligned}
E(p) &= \int_{\underline{p}}^{\tilde{p}} p dH(p) \\
&= pH(p)\,|_{\underline{p}}^{\tilde{p}} - \int_{\underline{p}}^{\tilde{p}} H(p)\, dp \\
&= 1\tilde{p} - 0\underline{p} - \int_{\underline{p}}^{\tilde{p}} H(p)\, dp \\
&= \tilde{p} - c.
\end{aligned}$$

To summarize:

1. What Quentin is worried about is not a concern in this problem,
2. $Ep = \int_{\underline{p}}^{\tilde{p}} (1 - H(p))\, dp$ is incorrect, it should be $Ep = \int_{\underline{p}}^{\tilde{p}} (1 - H(p))\, dp - \underline{p}$,

3. $\tilde{p} = E(p) + c$ is correct.

Suggestions for modifications to the content:

1. Abandon the use of B. Only use $\tilde{p}$ (and probably mention why this is the case by a few words),

   (a) in the original paper, the authors use p^* to denote some exogenous reservation price (in our notation, B) and z to denote the endogenous reservation price (in our notation, $\tilde{p}$) of consumers in the noisy search setting, and then $\bar{p} = \min(p^*, z)$ is the effective reservation price. And it is assumed implicitly that consumers know that firm will not offer price higher than the effective reservation price. Therefore we know $z \geq \bar{p}$ implies $H(z) = 1$. So it is safe (and consistent with the original paper) to use z (in our case, $\tilde{p}$) as the upper bound for p.

2. Derive the relationship between c, $\tilde{p}$, and $E(p)$ from the original definition of unconditional expectation $E(p) = \int_{\underline{p}}^{\tilde{p}} p dH(p)$.

Comments in the book

Let $v(p)$ be the expected price including future search costs of a consumer who has already paid c, has offer p in hand, and is about to decide whether to accept or reject the offer. The Bellman equation is

this B = p tilde

$$v(p) = \min_{\text{accept,reject}} \left\{ p, c + \int_{\underline{p}}^{B} v(p')\, dH(p') \right\}. \tag{6.6.2}$$

The reservation price $\tilde{p}$ satisfies $v(\tilde{p}) = \tilde{p} = c + \int_{\underline{p}}^{\tilde{p}} p' dH(p')$, which implies[15]

$$c = \int_{\underline{p}}^{\tilde{p}} H(p)\, dp. \tag{6.6.3}$$

this holds when p tilde = B and p underbar = 0

Combining equation (6.6.3) with the formula $Ep = \int_{\underline{p}}^{\tilde{p}} (1 - H(p)) dp$ for the mean of a nonnegative random variable implies that the reservation price $\tilde{p}$ satisfies

this equality holds as long as p tilde = B, even if p underline != 0

$$\tilde{p} = c + Ep,$$